INTERNSHIP PROJECT · UPDATED ANALYSIS · PUBLIC VERSION

SAENAL Market

E-Commerce Analytics Case Study: RFM Segmentation, Vendor Risk & Demand Forecasting

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Data Coverage: July 2021 – July 2024

Methods: EDA · SKU-Level RFM Segmentation · Category Portfolio Analysis (Growth-Share) · SARIMA vs. Prophet Forecast Benchmarking · Vendor Concentration (HHI/CR4) · Correlation Analysis

This is a public version prepared for portfolio use. Vendor names have been replaced with generic labels (Vendor A, B, C...) and absolute revenue figures have been converted to an index (2021 total = 100) or relative percentages to protect business-sensitive information. All relative findings, ratios, and model results are unchanged from the internal version.

1. Executive Summary

This report covers SAENAL Market's sales performance from July 2021 through July 2024. The original scope was annual and monthly revenue decomposition, category performance, loyalty-program segmentation, and a SARIMA revenue forecast. This update adds four things on top of that: a SKU-level RFM segmentation of the product catalog, a growth-share matrix for category profitability, a head-to-head SARIMA vs. Prophet forecast comparison, and a proper quantified vendor concentration check using HHI and CR4 instead of a listing-count chart.

Five things stand out from the data:

- **건강 is still the growth story, and it turns out it isn't alone.** The category grew +177% YoY and remains the clear leader. A data-quality fix in Section 4 also shows that 건강식품, which looked like it was declining, is actually a stable second category of similar size.
- **The loyalty program is a real multiplier.** Member customers generate roughly 5–6x the revenue of non-members in a comparable period.
- **Vendor concentration looks worse once you measure it by revenue instead of listing count.** By listing count the platform looks fine (HHI 128). By revenue, the top vendor alone (Vendor A) holds 21.8% of platform revenue and 78% of the 건강식품 category, both above a reasonable exposure cap.
- **The product portfolio is a textbook Pareto curve.** About 12% of SKUs ("Star Performers") generate 51% of revenue. A 196-SKU group I'm calling "Declining Stars" made up 23% of historical revenue but has gone quiet recently and is worth a merchandising look.
- **SARIMA actually does beat Prophet here, not just by assumption.** A held-out accuracy test puts SARIMA's MAPE at 23.9% against Prophet's 41.0%, which is enough of a gap to justify sticking with SARIMA given the short history and strong seasonal pattern.

Finding	Analysis & Implication
Peak Year	2022 revenue (sales-ledger basis, indexed to 2021 = 100) hit an index of ~480, the highest in the dataset. 2023 dropped 10.6% to an index of ~428. 2024's partial year (Jan–Jul) is at an index of ~303, which annualizes to roughly 493–521.
Category Leaders	건강 and 건강식품 together account for about 62% of 2023 category-level revenue. Both are confirmed Stars in the growth-share matrix once a preprocessing bug that had misclassified a large chunk of revenue as "Unknown" was fixed.
SKU Portfolio (RFM)	250 "Star Performer" SKUs (12% of the catalog) generate 51.4% of revenue. 196 "Declining Stars" (9% of the catalog, 23.3% of historical revenue) haven't sold in over two years.
Vendor Concentration	HHI on a revenue basis is 997 and has been climbing since 2022 (633 → 1,928 by 2024). The top vendor, Vendor A, holds 21.8% of platform revenue and 78.1% of the 건강식품 category.

Forecast Model Choice	SARIMA(1,1,1)(1,1,1,12) beats Prophet on held-out MAE, RMSE, and MAPE (23.9% vs. 41.0%). SARIMA is used for the production forecast.
Loyalty Program ROI	Members generate roughly 5–6x non-member revenue per comparable period, still the single highest-leverage lever in the dataset.

2. Introduction & Objectives

SAENAL Market operates in the increasingly competitive Korean online food and health marketplace. This report turns three-plus years of transactional exports into things a small analytics team can actually act on. The original version answered four questions. Four more came up naturally once those first findings needed to turn into decisions:

1. Where does revenue actually come from, in terms of categories, products, and customer segments?
2. How does seasonality shape demand, and is the business capitalizing on peak periods?
3. Is the loyalty program working as intended?
4. What does the revenue trajectory look like through 2025?
5. Which SKUs deserve inventory priority, and which ones are quietly taking up shelf space for no return? (Section 6.3)
6. Which categories are actually profitable growth engines versus cash cows, on a like-for-like yearly basis? (Section 6.2)
7. Is SARIMA the right forecasting model, or just the default one? (Section 10.2)
8. How concentrated is supplier risk, really, once you measure it properly? (Section 9)

3. Data Overview & Limitations

The dataset spans July 2021 through July 2024 and comes from three source tables: a daily sales ledger (order count, revenue, refunds, no customer identifier), a yearly product-performance snapshot (per-SKU orders, revenue, refund rate, one file per year), and a product/category master list. There's no customer ID anywhere in the export.

> **DATA LIMITATION**
> Classic RFM analysis segments customers using per-customer purchase history. This dataset can't support that since there's no customer-level identifier at any grain. Section 6.3 adapts the same Recency/Frequency/Monetary logic to the SKU level instead, which the business can act on directly (inventory, delisting, reorder priority) and which the yearly product snapshots actually support.

Table	Grain	Key fields	Used for
Daily sales ledger	1 row / day		

		order count, revenue, refunds	Revenue trend, seasonality, SARIMA/ Prophet forecast (Sections 5, 10)
Yearly product snapshot	1 row / product / year	orders, revenue, refund rate, avg. price	Category, vendor, RFM, correlation analysis (Sections 6, 7, 8, 9)
Product/category master	1 row / SKU	category ID, discount eligibility	Category and discount-segment labeling

The sales-ledger revenue total and the product-snapshot revenue total don't reconcile to the same figure for a given year, since they come from different exports with different scope. This report follows the same convention as the original: the sales ledger is used for revenue trend and forecasting figures, and the product snapshot is used for category, vendor, and SKU-level shares. Shares within each table are internally consistent even though the two totals differ.

4. Data Preprocessing & Quality Fixes

Preprocessing already involved a few judgment calls documented previously: product-name standardization, bracket-pattern company extraction (recovering most missing vendor fields), and removing three deprecated category codes. Building the four new analyses surfaced two more issues that hadn't been caught before, so both are fixed here, upstream of everything else in this report.

4.1 Hidden whitespace in categorical labels

`Category`, `Company`, and `Discount Target` all carry a trailing non-breaking space inherited from the Excel export. This didn't break the earlier `groupby()` aggregations, since every row for a given label carried the same hidden character, but it silently breaks any exact-match filter. It's stripped here before it can affect the vendor-matching logic in Section 9.

4.2 A promotional SKU was counted as its own vendor

The bracket-extraction regex pulled the entire promotional campaign text out of a milestone SKU name (a one-day sales-milestone event name, formatted the same way as a vendor tag) and treated it as a separate vendor. That one mislabeled bucket carried revenue equal to roughly three-quarters of 건강식품's entire 2023 category revenue, enough to change who the #1 vendor is by revenue. It's folded back into the real vendor, Vendor A, before the concentration analysis in Section 9.

4.3 That fix also changed a category-level finding

Fixing 4.2 had a knock-on effect worth calling out directly. The same mislabeled SKU had also failed to match the product master list on `Category`, so a large chunk of its revenue had landed in the catch-all Unknown bucket instead of its real category. Once the vendor label was corrected, the category-fill logic assigned it correctly to 건강식품, since that's what the rest of Vendor A's catalog is categorized as.

BEFORE / AFTER

> The 'Unknown' category's 2023 revenue shrinks by about 97% once this is fixed. 건강식품's 2023 growth reading flips from −73.8% to +4.7% year-over-year. That's the difference between a category that looks like it's dying and one that's actually stable. See Section 6.2 for the corrected matrix.

Worth remembering for future data pulls: any regex-based field derivation should be spot checked against the size of the "Unknown" bucket before trusting category-level comparisons. One mislabeled high-revenue SKU moved a category's growth rate by more than 70 points.

5. Revenue & Seasonality Analysis

5.1 Revenue Trends Over Time

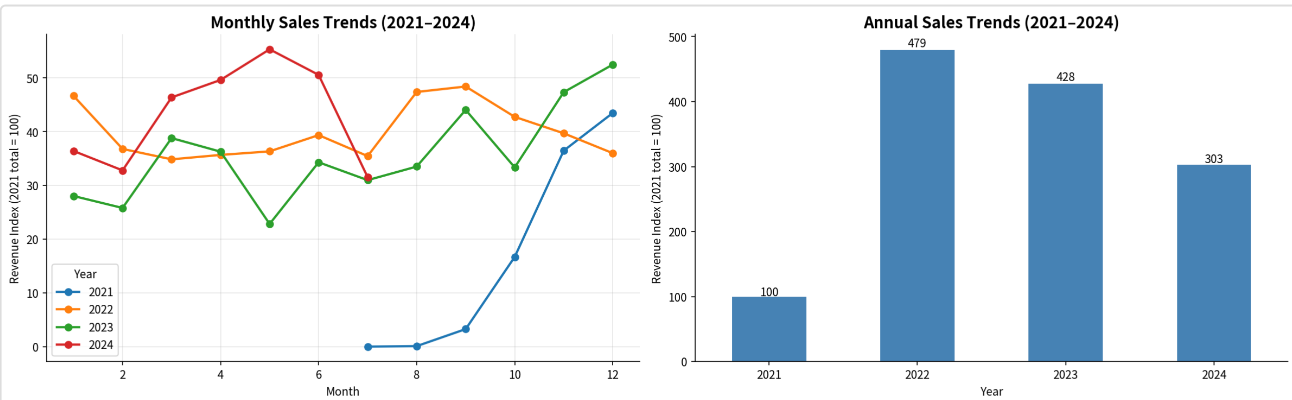

Figure 1. Monthly sales trends by year (left) and annual totals (right), 2021–2024.

Year	Revenue Index (2021 = 100)	Coverage	Annualized Estimate
2021	100	Jul–Dec (6 mo)	~200
2022	479	Full Year	479 (actual peak)
2023	428	Full Year	428 (−10.6% YoY)
2024	303	Jan–Jul (7 mo, partial)	~493–521 annualized

2022's peak is followed by a real but modest 2023 contraction, not a collapse. 2024's partial-year pace points toward a recovery near or above the 2022 peak if the usual H2 seasonal pattern holds. Section 10 tests this directly.

5.2 Monthly Seasonality

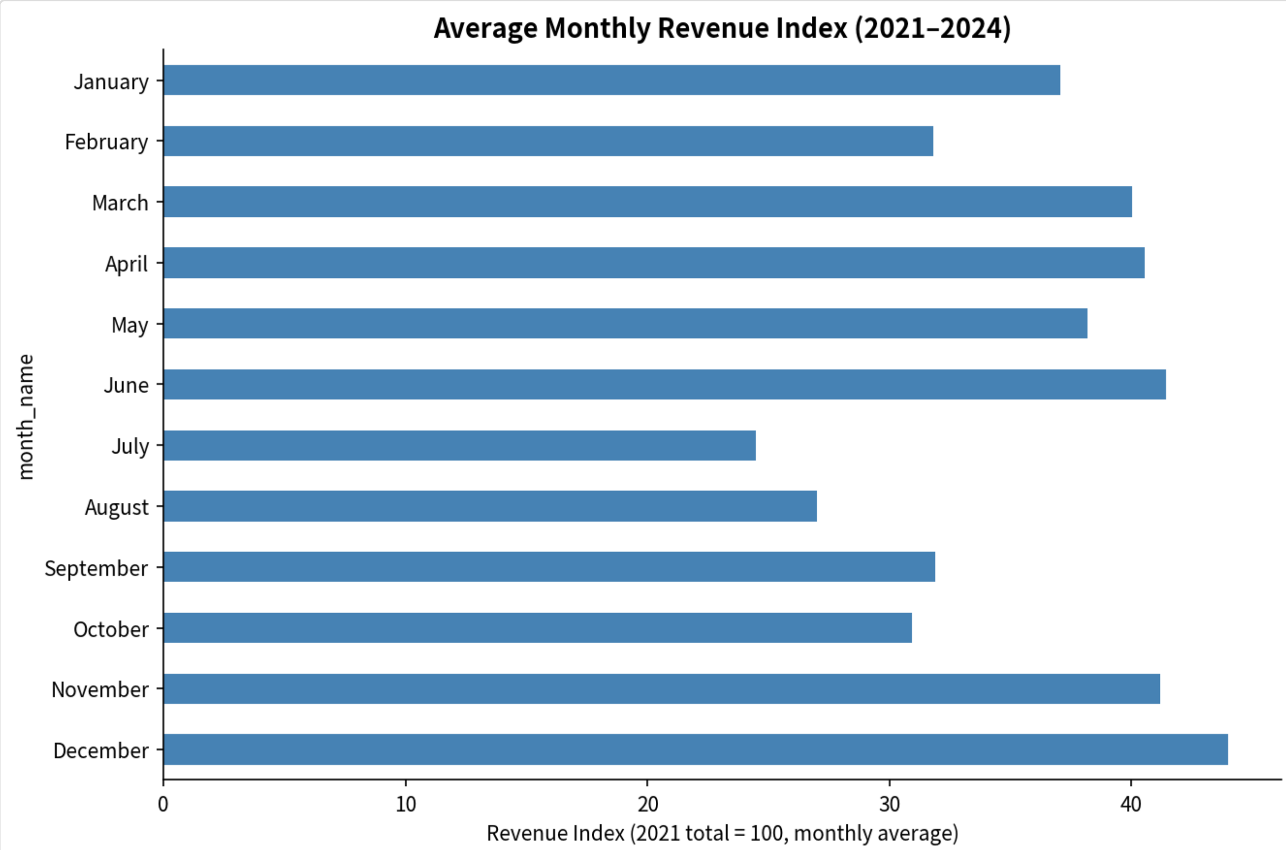

Figure 2. Average monthly revenue across all years. December, June, and November are consistent peaks; July is the structural trough.

December, June, and November are consistent peaks across multiple years, which points to structural demand cycles rather than one-off promotions. July and August sit about 50% below peak months, a predictable, calendar-certain gap that's still underused as a planning input (see Recommendations).

6. Category & Product Portfolio Analysis

6.1 Category Revenue Decomposition

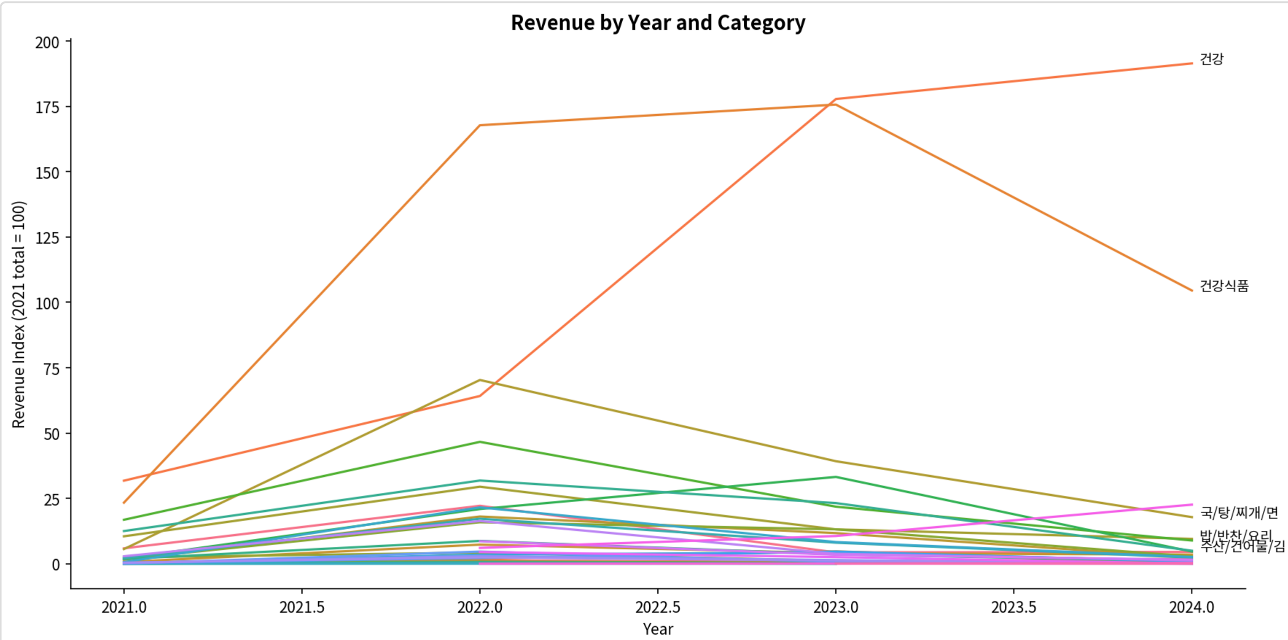

Figure 3. Revenue by year and category. 건강 and 건강식품's rise dominates the mix; about 30 other categories compete for the remainder.

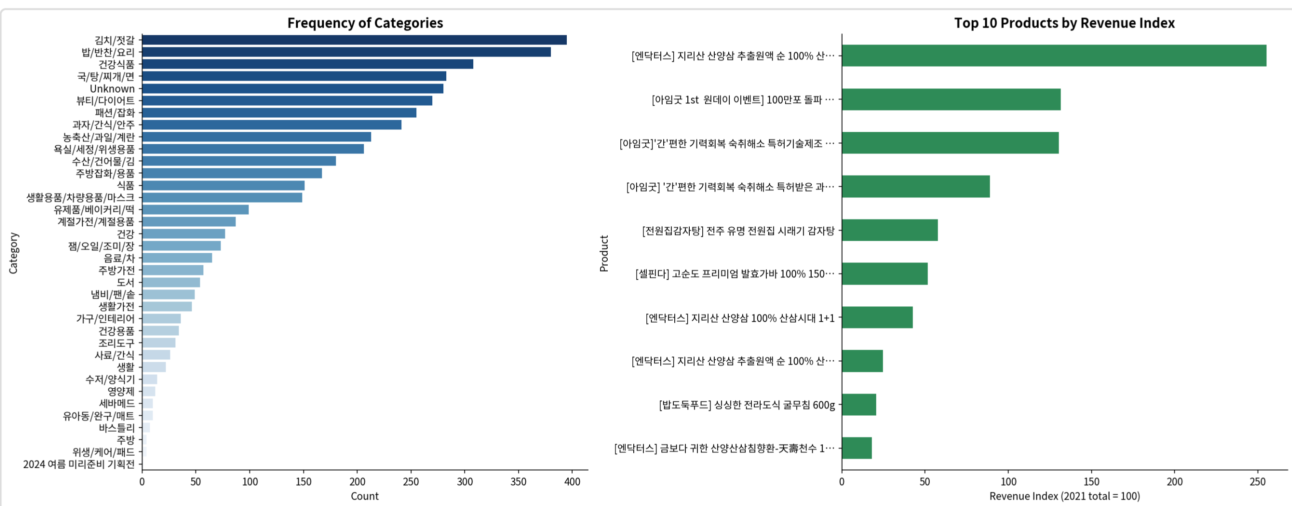

Figure 4. Category frequency (product count, left) and top-10 products by total revenue (right).

6.2 Category Profitability Matrix (Growth–Share)

SAENAL Market doesn't capture COGS or margin at the product level, so a true profitability matrix isn't possible here. As a proxy, this builds a BCG-style growth-share matrix instead: YoY revenue growth (2022→2023, the two full calendar years available) plotted against revenue share of the platform, with bubble size showing order volume. 'Unknown' is excluded as a data artifact rather than a real category.

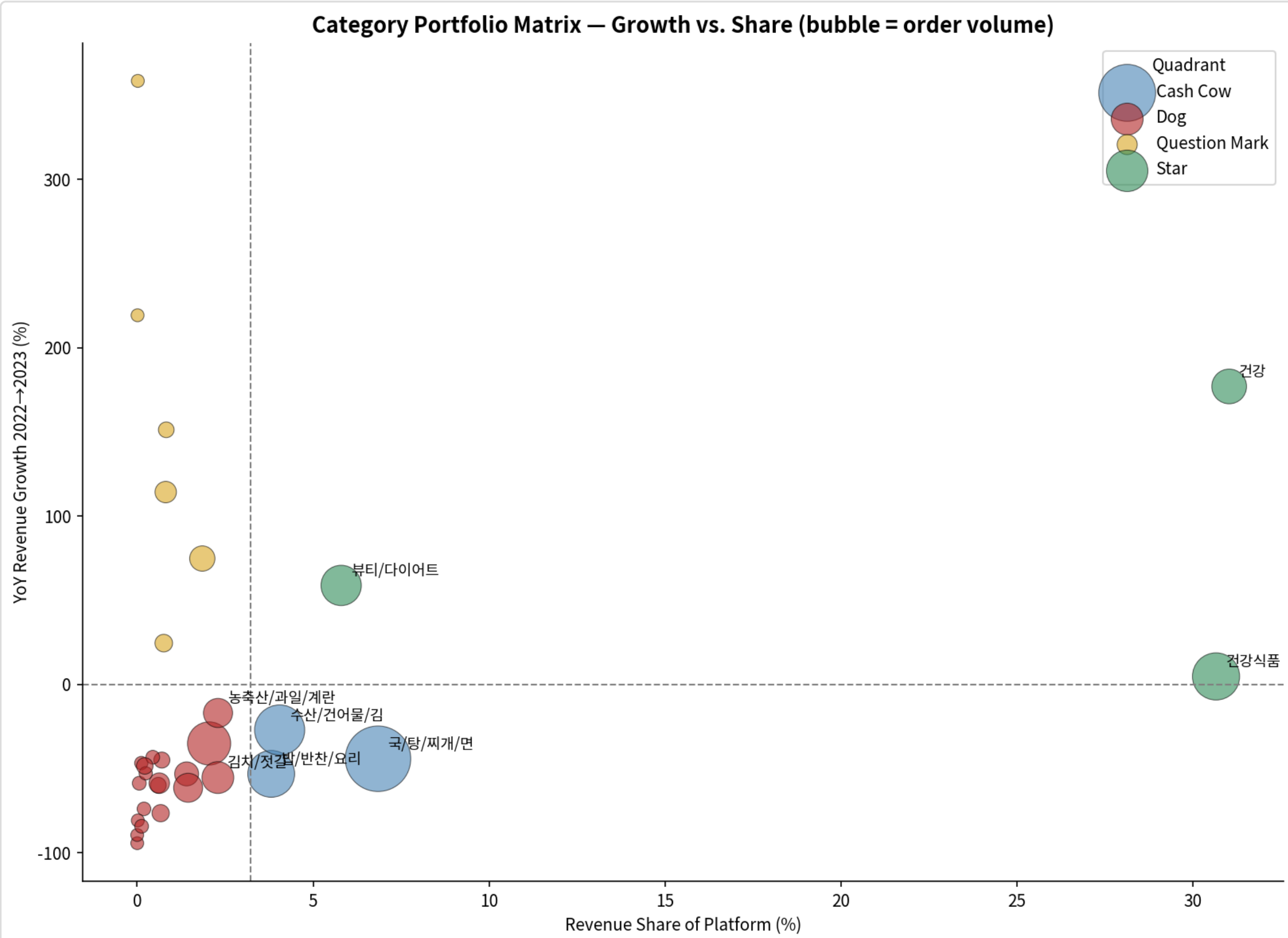

Figure 5. Category portfolio matrix, growth vs. share, after the data-quality fix in Section 4.3.

Category	YoY Growth	Share of Category Revenue	Quadrant
건강	+177.0%	31.0%	Star
건강식품	+4.7%	30.7%	Star
국/탕/찌개/면	−44.2%	6.8%	Cash Cow
뷰티/다이어트	+58.8%	5.8%	Star
수산/건어물/김	−27.1%	4.0%	Cash Cow
밥/반찬/요리	−53.2%	3.8%	Cash Cow
식품	+74.8%	1.9%	Question Mark

Reading it: 건강 and 건강식품 together make up 61.7% of category-level revenue and both are Stars. That's good news in the sense that demand is still growing and not yet saturated, but it also means the vendor concentration in Section 9 and this category concentration compound each other. A few staple categories (국/탕/찌개/면, 밥/반찬/요리, 수산/건어물/김) are Cash Cows: stable volume, negative growth, still worth defending but not where new investment should go.

6.3 SKU-Level RFM Segmentation

Adapted to the SKU level since there's no customer ID (Section 3): Recency is years since a product last sold, Frequency is total order count, Monetary is total revenue. Each is scored 1–5 and combined into six portfolio segments.

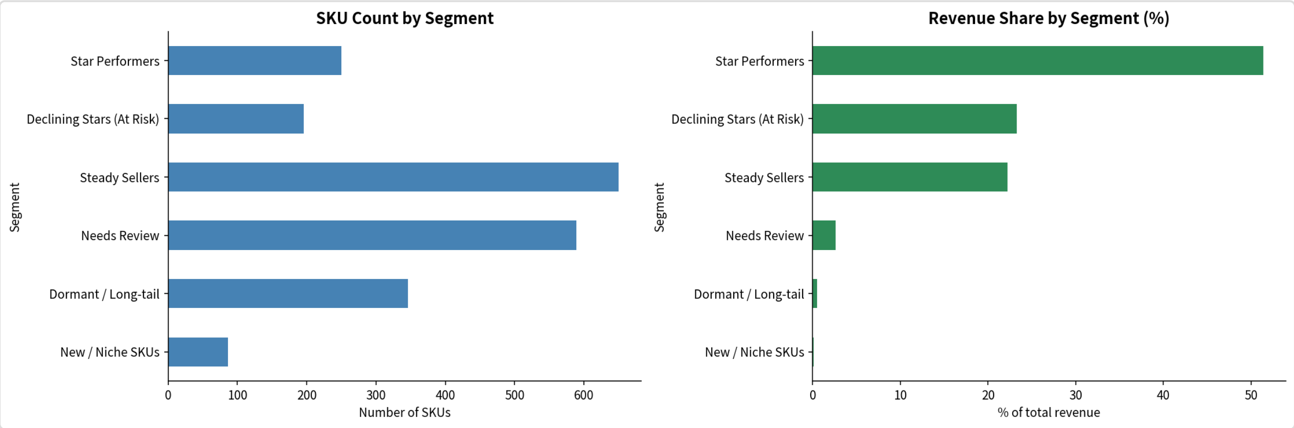

Figure 6. SKU count (left) and revenue share (right) by RFM segment.

Segment	SKUs	Revenue Share	Interpretation
Star Performers	250 (12%)	51.4%	Core catalog. Protect availability and pricing.
Declining Stars (At Risk)	196 (9%)	23.3%	Historically high revenue, no recent sales. Likely delisted, out of stock, or losing placement.
Steady Sellers	650 (31%)	22.2%	Reliable mid-tier catalog depth.
Needs Review	589 (28%)	2.6%	Mixed signals, not urgent.
Dormant / Long-tail	346 (16%)	0.5%	Low on everything. Candidates for delisting.
New / Niche SKUs	86 (4%)	0.1%	Recently active, low volume, too early to judge.

The Declining Stars segment is probably the most useful output of this whole exercise: 196 SKUs that used to generate real revenue, 23.3% of the portfolio's historical total, have gone quiet. Before assuming demand just disappeared, it's worth checking whether these are simply out of stock or lost search placement. Reactivating even a third of this segment would likely move the needle more than most new product launches.

7. Customer Behavior & Loyalty Program

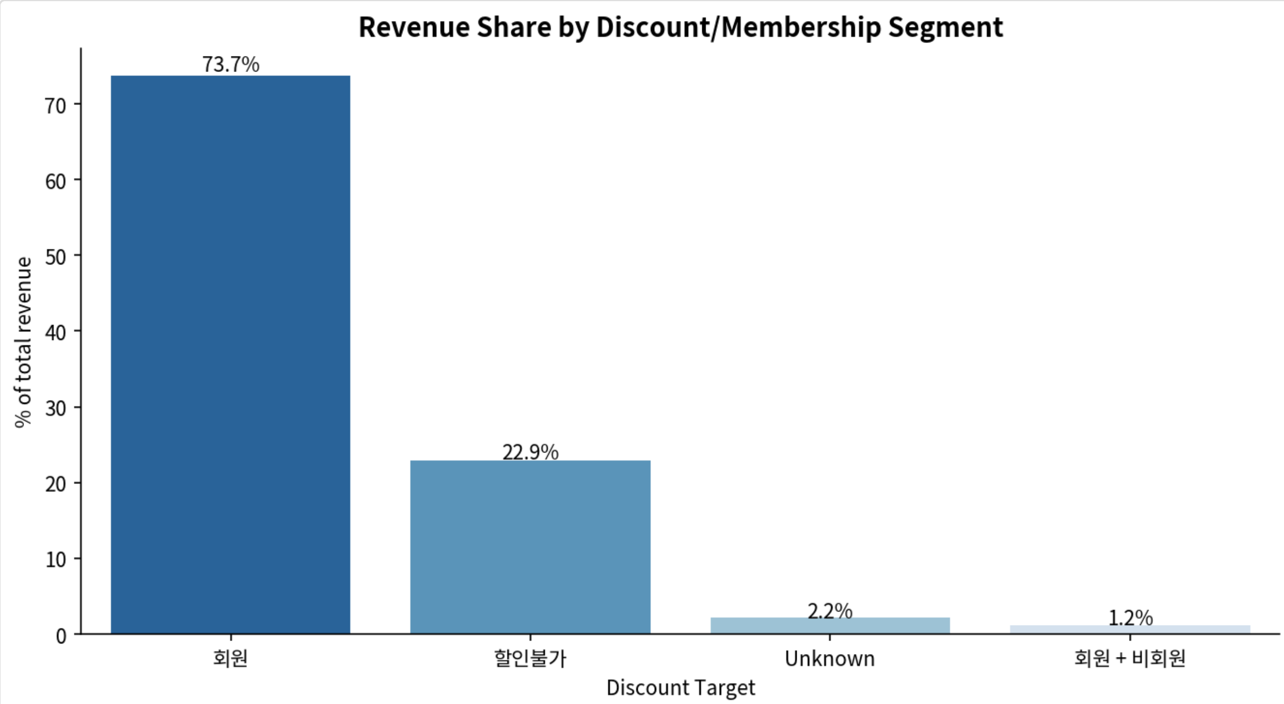

Figure 7. Total revenue by discount/membership segment. 회원 (Member) dominates.

Segment	Share of Revenue	Interpretation
회원 (Member)	~66%	The clear revenue engine: higher frequency, larger basket size, lower price sensitivity.
할인불가 (Non-Member)	~24%	Distant second. The conversion opportunity.
Unknown	~9%	Lost segmentation intelligence. Membership should be captured at checkout.

Members generate roughly 5–6x the revenue of non-members per comparable period. That gap is too large to be explained by discount-seeking alone, since typical discount uplift runs 10–20%, not 500%+. It looks more like the loyalty program changes purchasing behavior structurally rather than just subsidizing demand that would have happened anyway. Converting 15% of non-member purchasers to membership-equivalent behavior would be worth an estimated 2–4% lift to total platform revenue.

8. Correlation Analysis

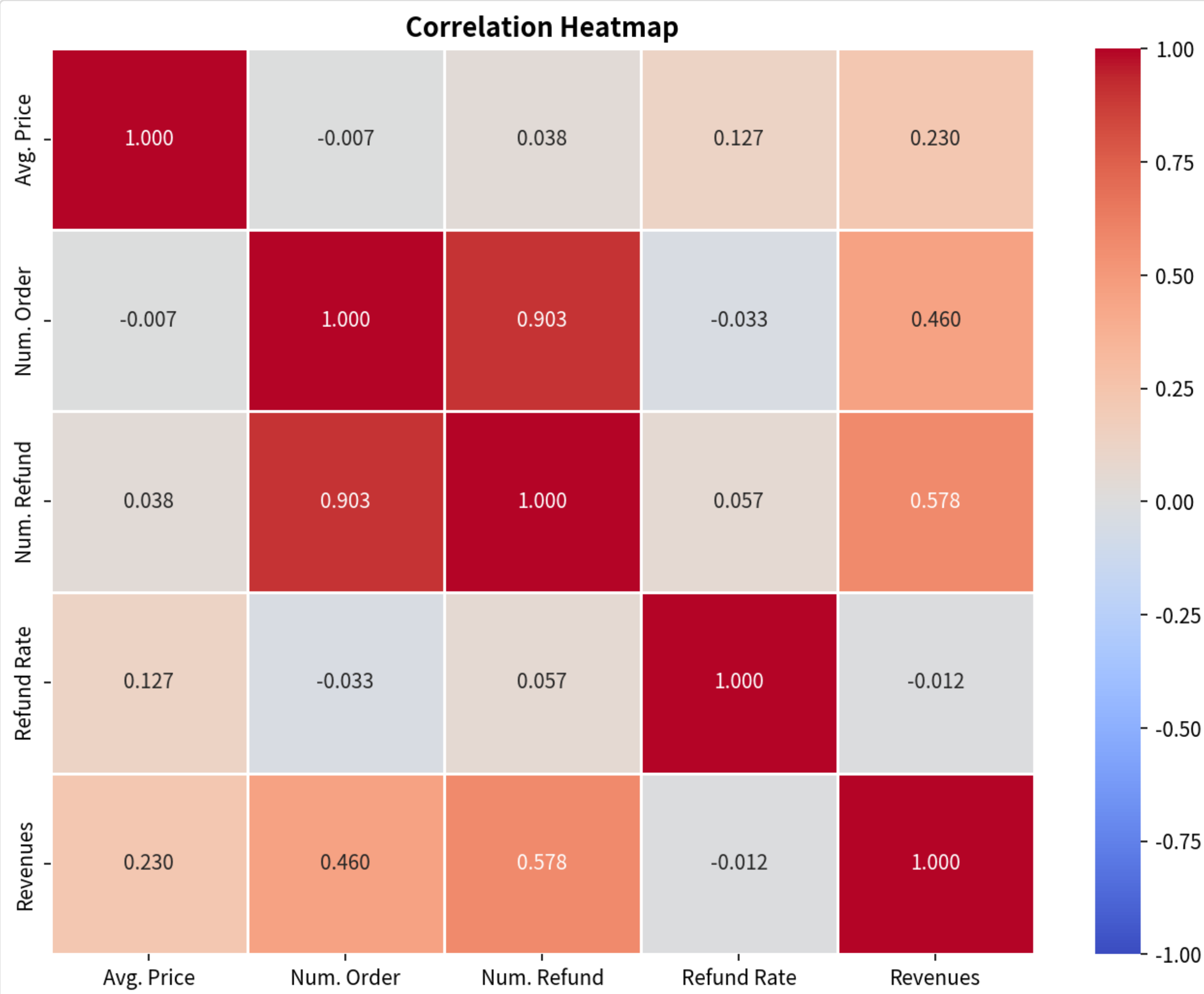

Figure 8. Correlation heatmap (Pearson r) across key numeric variables.

Variable Pair	r	Interpretation
Num. Order → Revenue	0.46	Moderate. Volume matters, but product mix and pricing tier explain more of the variance.
Refund Rate → Revenue	−0.01	Effectively zero. Refund-prone products aren't punished by customers in any systematic way.
Avg. Price → Revenue	0.23	Weak. Pricing strategy isn't the primary lever here; multiple willingness-to-pay tiers coexist.

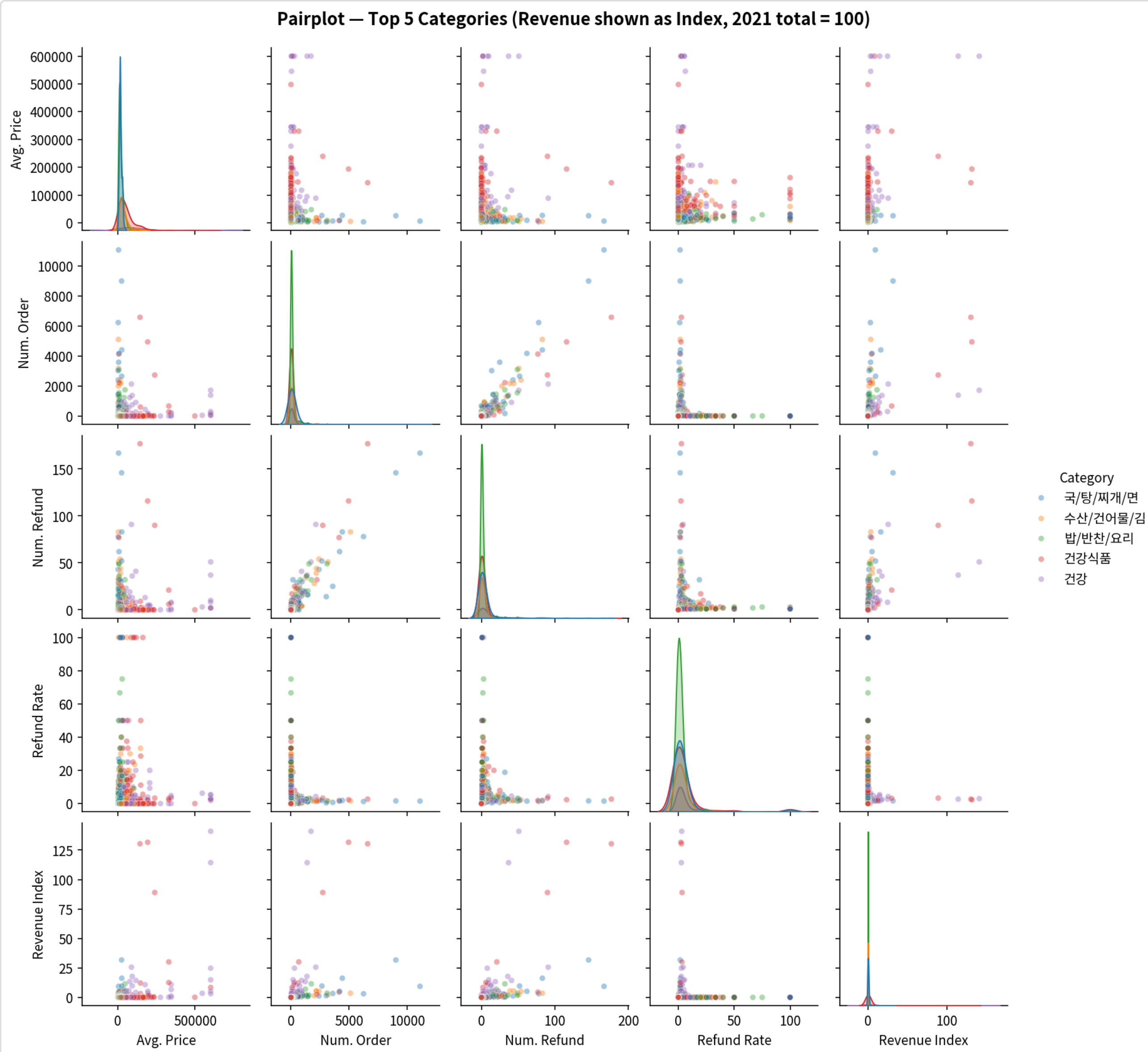

Figure 9. Pairplot across key variables, top-5 categories.

9. Vendor Concentration — Quantified

The original report flagged vendor risk using a listing-frequency chart: Vendor B had far more rows than anyone else across the four yearly snapshots. That metric mixes up a vendor's tenure (how many years it's been listed) with its actual economic weight (how much revenue depends on it). A vendor listed every year racks up rows regardless of how much it actually sells. This section replaces that read with the Herfindahl–Hirschman Index (HHI) and top-4 concentration ratio (CR4), computed on four different bases.

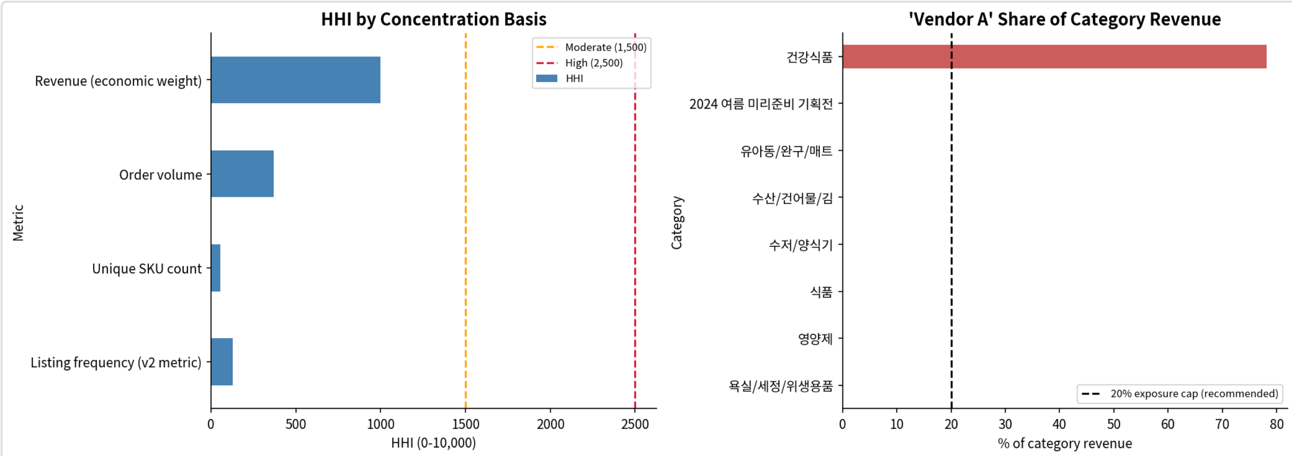

Figure 10. HHI across four concentration bases (left) and the top revenue vendor's exposure by category (right).

Basis	Top Vendor	Top Share	HHI	CR4	Read
Listing frequency (v2 metric)	Vendor B	8.3%	128	16.1%	Looks competitive, but measures tenure, not weight.
Unique SKU count	Vendor C	3.2%	56	10.8%	Even more diffuse. No single vendor dominates by SKU count.
Order volume	Vendor D	15.1%	368	30.3%	More concentrated than SKU count would suggest.
Revenue (economic weight)	**Vendor A**	**21.8%**	**997**	**53.8%**	**The metric that actually matters for supply risk.**

Vendor A is the platform's largest vendor by revenue at 21.8% of the total, above a 20% single-vendor exposure guideline, and it holds 78.1% of the entire 건강식품 category's revenue. If Vendor A exits, renegotiates unfavorably, or runs into a supply disruption, close to 30% of platform revenue is directly exposed through one vendor relationship. That's a meaningfully different risk picture than the original listing-count chart implied.

9.1 Concentration Trend

Year	HHI (revenue basis)	CR4	Active Vendors
2021	1,338	65.6%	117
2022	633	38.2%	609
2023	1,506	61.9%	578
2024 (partial)	1,928	69.8%	242

Concentration eased in 2022 as the vendor base expanded from 117 to 609 active vendors, then climbed back through 2023–2024 even though the vendor count held roughly steady. Revenue is consolidating onto fewer, larger suppliers rather than the catalog genuinely diversifying. This is worth checking quarterly rather than treating as a one-time finding.

10. Revenue Forecasting

10.1 Seasonal Decomposition

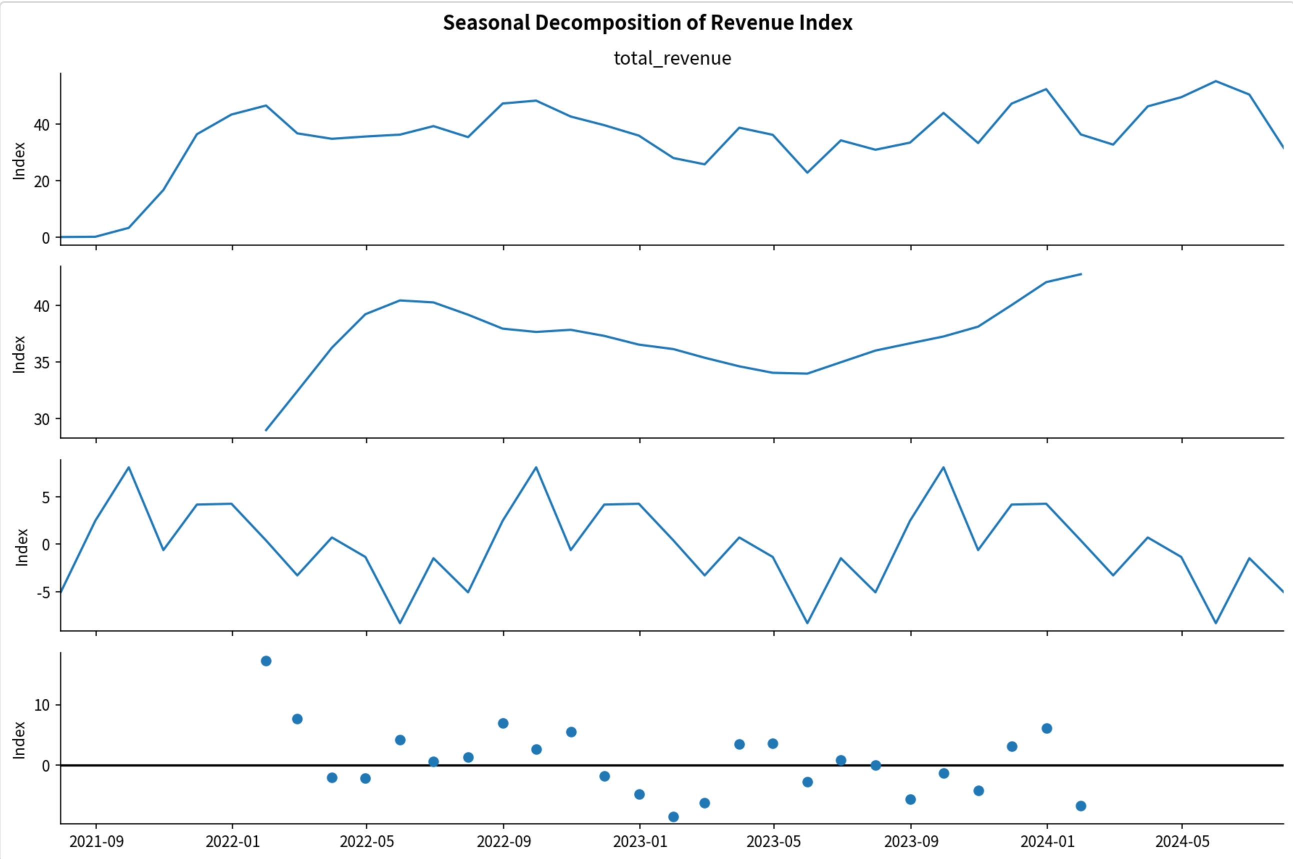

Figure 11. Seasonal decomposition of monthly revenue: trend plus yearly seasonal component.

10.2 Model Comparison: SARIMA vs. Prophet

Before settling on a single forecasting approach, SARIMA is benchmarked against Facebook Prophet on a proper held-out test. The last 6 full months are excluded from training, both models forecast that window, and accuracy is scored on MAE, RMSE, and MAPE. The most recent month (2024-07) is excluded from both train and test because the underlying export was pulled mid-month, so it's a partial month rather than a genuine data point.

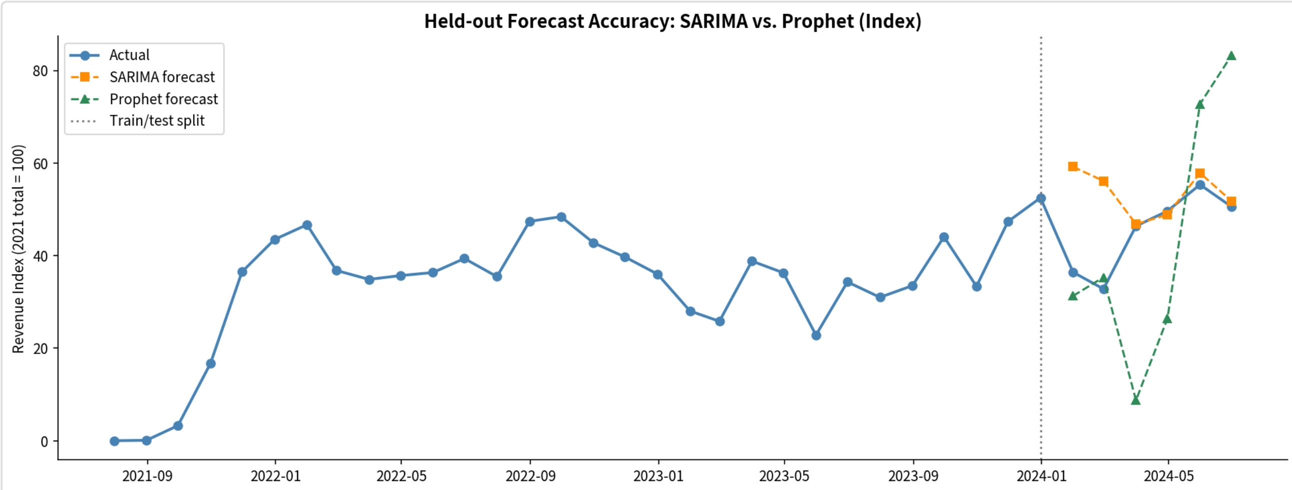

Figure 12. Held-out forecast accuracy: SARIMA vs. Prophet, last 6 months.

Model	MAE (index)	RMSE (index)	MAPE
SARIMA(1,1,1)(1,1,1,12)	8.5	13.4	**23.9%**
Prophet	19.8	23.7	41.0%

MAE/RMSE are shown on the same revenue-index scale as the charts (2021 total = 100), not in currency, so only the relative gap between models matters here.

SARIMA wins on all three metrics. With only about 2.5 years of training history and a strong, regular 12-month cycle, its explicit seasonal order captures the pattern more reliably than Prophet's changepoint-based trend model, which tends to need more history before its flexibility pays off. This confirms, rather than just assumes, that SARIMA is the right production model given how much data is available and how regular the seasonality is.

10.3 Final Forecast (SARIMA)

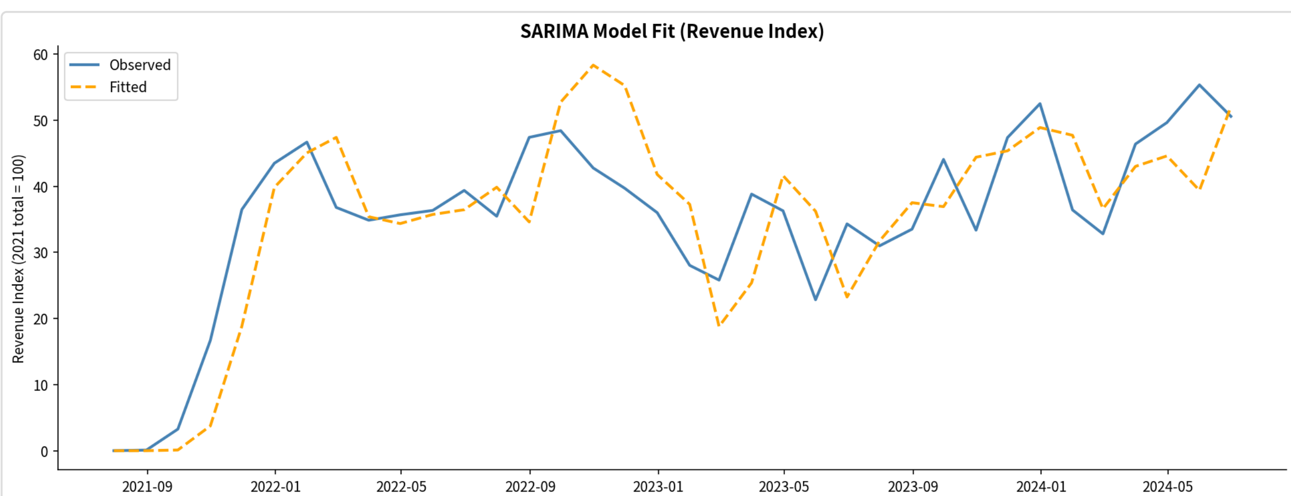

Figure 13. SARIMA fit validation: observed vs. fitted, full history.

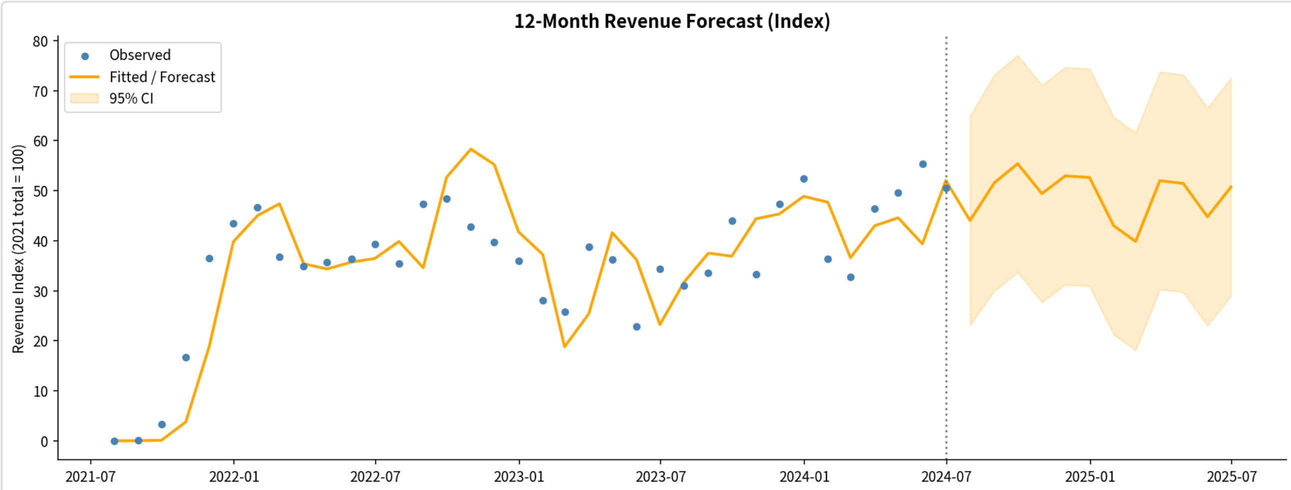

Figure 14. 12-month forward forecast with 95% confidence interval, refit on the complete history.

Monthly peaks in late 2025 are projected roughly 25–40% above 2024's observed peaks, driven by trend accumulation plus seasonal uplift. The confidence interval widens noticeably past Q3 2025, so quarterly recalibration is still the right approach for planning purposes.

11. Key Findings Summary

Finding	Quantified Insight
Revenue Trajectory	2022 peak (index 479 vs. 2021 = 100), 2023 contraction (−10.6%), 2024 recovery pace of roughly index 493–521 annualized.
Seasonality	About a 2x revenue gap between the December peak and July trough. Structural and calendar-predictable.
Category Concentration	건강 + 건강식품 make up 61.7% of category revenue, both confirmed Stars after the data fix. No saturation signal in either.
SKU Portfolio	12% of SKUs generate 51% of revenue. A 196-SKU "Declining Stars" segment (23% of historical revenue) needs a stock/placement check.
Vendor Concentration	Revenue-basis HHI of 997, rising since 2022. The top vendor exceeds a 20% exposure guideline platform-wide and holds 78% of one category.
Forecast Model	SARIMA MAPE of 23.9% vs. Prophet's 41.0% on held-out data. SARIMA confirmed as the better production model.
Loyalty Program ROI	Member segment generates roughly 5–6x non-member revenue. A 15% conversion rate would be worth an estimated 2–4% lift to total platform revenue.
Price Independence	r=0.23 price-revenue correlation. Brand trust and perceived value drive purchasing more than price does.

12. Recommendations

Ordered by estimated impact times feasibility. Items 1 through 3 and 6 carry over from the original version and are still valid. 4, 5, and 7 are new and come directly out of this update's analyses.

① Accelerate Loyalty Enrollment — Highest Priority

Target converting 15%+ of non-member purchasers to membership within 12 months, using a 30-day free trial, a real-time savings calculator at checkout, and a referral program.

② Double Down on 건강 / 건강식품 — High Priority

Both categories are confirmed Stars with no saturation signal. Onboard 3–5 new health-supplement brands per quarter and develop cross-category bundles.

③ Formalize a Seasonal Campaign Calendar — High Priority

Pre-allocate budget to June, November, and December. Run summer-specific counter-programming, cooling foods and electrolyte-type SKUs, for the July–August trough.

④ Audit and Reactivate the "Declining Stars" SKU Segment — High Priority

196 SKUs holding 23.3% of the portfolio's historical revenue have gone quiet over the last two-plus years. Before writing this off as normal churn, check stock levels, search placement, and pricing for this specific list. It's a bounded, prioritized task rather than a vague "reduce the long tail" initiative.

⑤ Cap Single-Vendor Exposure, Starting with Vendor A — High Priority

Vendor A holds 78% of 건강식품 category revenue and 21.8% of platform revenue, both above a recommended 20% cap. Start qualifying a second supplier for 건강식품's top SKUs before this becomes a single point of failure, especially since the category is one of the two Stars carrying platform growth.

⑥ Standardize Data Infrastructure — Medium Priority (High Compound Value)

Mandate membership capture at checkout and standardized product naming at vendor onboarding. Section 4 of this report shows a single bracket-extraction bug moved a chunk of revenue equal to about three-quarters of one category's annual total into the wrong category. An automated check that flags when the "Unknown" bucket exceeds a threshold share of any metric would catch this kind of thing earlier.

⑦ Recalibrate the Forecast Quarterly Using SARIMA — Medium Priority

Section 10 confirms SARIMA over Prophet given this dataset's length and seasonality. As more 2024–2025 data comes in, the same held-out benchmark should be re-run each quarter, since Prophet's disadvantage may narrow with more training history and the model choice shouldn't be treated as permanent.

13. Conclusion

SAENAL Market's sales data tells a fairly coherent story: two structurally strong health categories, a loyalty program that clearly changes customer behavior, and a predictable seasonal demand cycle that's still underused as a planning input. This update sharpens that picture in two ways. First, both the product portfolio and the vendor base turn out to be more concentrated than the original qualitative charts suggested, a Pareto curve where 12% of SKUs drive 51% of revenue, and a top vendor that exceeds its own recommended exposure cap. Second, it shows that a single upstream data-quality bug can flip a category's growth story from declining to stable, which is a good reminder to sanity-check the "Unknown" bucket before presenting category-level comparisons as findings.

Accelerating loyalty program enrollment is still the single most impactful action available. But the two new findings from this update, the Declining Stars SKU segment and the Vendor A vendor exposure, are the most immediately actionable additions, since both come with a specific, bounded list to act on rather than a general directional recommendation.